

Multivariable calculus and SVM: notes and theory

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Abstract

Some notes about multivariate analysis in the direction of SVM and machine learning. Nothing serious.

Keywords: Linear algebra, multivariate analysis, SVM, kernel-trick.

1 The basics

In principio erat Verbum. The goal of this document is to give a short introduction and an intuition to Supported Vector Machine (SVM).

1.1 Scalar product

Despite the ground reasons for which scalar product can be used, here we check that everything is reasonable. Consider two vectors in $\mathbb{R}^2$, $a = (a_x, a_y)$ and $b = (b_x, b_y)$ and consider the usual scalar product in $\mathbb{R}^2$. Called θ the angle between a and b and γ the angle between b and the x -axes we have:

$$\begin{aligned}a_x &= \|a\| \cos(\gamma + \theta), \\a_y &= \|a\| \sin(\gamma + \theta), \\b_x &= \|b\| \cos(\gamma), \\b_y &= \|b\| \sin(\gamma).\end{aligned}$$

Now:

$$\langle a, b \rangle = a_x b_x + a_y b_y = \|a\| \|b\| [\cos(\theta) (\cos^s \gamma + \sin^2 \gamma)] = \|a\| \|b\| \cos \theta.$$

Then product scalar is connected to the angle between vectors. In a more general space is then reasonable to define the notion of angle θ starting from a scalar product.

$$\cos \theta = \frac{\langle a, b \rangle}{\|a\| \|b\|}.$$

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1.2 Stuff about planes

What is the definition of a plane? What is a plane? Once that you have defined the scalar product and the notion of angle, you can define a plane as the set of points orthogonal to a given vector: in formulas:

$$\langle w, x \rangle + b = 0,$$

where b somehow plays the role of intercept.

The concept of distance plays an important role in $\mathbb{R}^n$. Given a point x_0 how can you compute the distance of that point from a plane with normal vector w ?

Draw a plot to convince yourself of the following: take x_H which belongs to the plane, $\langle w, x_H \rangle + b = 0$ and set $v = x_0 - x_H$. This is vector linking the point x_0 to a general point in the plane. The distance d is given by the projection of v on w , and the projection is:

$$d = \left| \frac{\langle w, v \rangle}{\|w\|} \right| = \left| \frac{\langle w, x_0 - x_H \rangle}{\|w\|} \right| = \left| \frac{\langle w, x_0 \rangle - \langle w, x_H \rangle}{\|w\|} \right| = \frac{|\langle w, x_0 \rangle + b|}{\|w\|}.$$

What is b , then? Consider $x_0 = \mathbf{0}$ then $d = \frac{|b|}{\|w\|}$. Hence b regulates somehow the distance from the origin.

Now focus on the following problem. Given two planes, which are parallel, which is the distance between them? This stuff is useful in the context of linear SVM. Two parallel planes have the same w , but different scalar b .

$$\langle w, x \rangle + b = 0, \langle w, x \rangle + b' = 0$$

Consider x'_0 in the second plane na compute the distance from the first one:

$$d = \frac{|\langle w, x'_0 \rangle + b|}{\|w\|} = \frac{|b - b'|}{\|w\|}.$$

Then the distance between two parallel planes is proportional to the difference between b and b' .

1.3 Directional derivatives and gradient

Definition 1.1. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be differentiable at $x_0 \in \mathbb{R}^n$ if there exists a linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}$ such that for $\|u\| \rightarrow 0$ we have that:

$$f(x_0 + u) = f(x_0) + L(u) + o(\|u\|).$$

Is there an explicit expression for this operator L ? Consider a particular direction for $u = e_i h$, for h scalar and e_i the canonical basis of $\mathbb{R}^n$. Then:

$$\begin{aligned} f(x_0 + h e_i) &= f(x_0) + L(h e_i) + o(\|h e_i\|) \\ &= f(x_0) + h L(e_i) + o(|h|) \end{aligned}$$

Then, taking the limit for $h \rightarrow 0$:

$$L(e_i) = \lim_{h \rightarrow 0} \frac{f(x_0 + he_i) - f(x_0)}{h} + \frac{o(h)}{h} = \frac{\partial f(x)}{\partial x_i} \Big|_{x=x_0} = \frac{\partial f(x_0)}{\partial x_i}$$

Consider a generic $u \in \mathbb{R}^n$ and the usual $x_0 \in \mathbb{R}^n$, then:

$$\begin{aligned} L(u) &= L\left(\sum_{i=1}^n u_i e_i\right) = \sum_{i=1}^n u_i L(e_i) = \sum_{i=1}^n u_i \frac{\partial f(x_0)}{\partial x_i} \\ &= u \cdot \left(\frac{\partial f(x_0)}{\partial x_1}, \dots, \frac{\partial f(x_0)}{\partial x_n}\right) = u \cdot \nabla f(x_0), \end{aligned}$$

Then the linear operator for differentiable functions from $\mathbb{R}^n$ to $\mathbb{R}$ is the gradient, the vector of partial derivatives valuated at x_0 .

Now consider the general directional derivative along the direction u , with $\|u\| = 1$:

$$D_u f(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + hu) - f(x_0)}{h} = \lim_{h \rightarrow 0} \frac{\nabla f(x_0) \cdot u + o(\|u\|h)}{h} = \nabla f(x_0) \cdot u.$$

Now if I ask: at x_0 what is the direction in which the increment is maximum (for small step h) I am asking myself which is the u which maximizes: $D_u f(x_0) = \|\nabla f(x_0)\| \|u\| \cos \theta$, where θ is the angle between the gradient and the vector u I have to choose. This quantity is maximum if $\cos \theta = 1$, namely if they are parallel, namely if I move parallel to the direction of the gradient.

It trivially follows that in order to minimize a function, you can move in the opposite direction of the gradient.

2 Lagrangian multipliers

In this section we briefly recover the Lagrangian method for minimization. We give an example and we discuss about the theory behind it.

2.1 Intuition

Assume you have a function $f(x, y)$ and that you want to find its extreme points but not on the whole $\mathbb{R}^2$ but only along a curve, namely subject on a given constraint of the form $g(x, y) = 0$.

In this case for example you may have $f(x, y) = xy + 1$ and $g(x, y) = x^2 + y^2 - 1$. Hence you are looking for extrema points on a circle centered in the origin.

Recall what a contour is: it is the set of all points (x, y) such that satisfies the relation $f(x, y) = c$, namely all those points for which the “height” of the function $z = f(x, y)$ is constant.

The crucial fact is that extrema occurs when contours are tangent to the constrain. This has to be proved.

Since they are tangent, they share the same tangent plane, which means that the gradient are parallel, namely:

$$\nabla f = \lambda \nabla g.$$

Moreover you have to request to stay in the constrain, namely $g(x, y) = 0$. This mean that you have to solve the following problem:

$$\begin{cases} \nabla f = \lambda \nabla g, \\ g(x, y) = 0. \end{cases}$$

You have tree equations, three unknown, hence the system is determined. Let's look at one example.

Example 2.1. Consider $f(x, y) = xy + 1$ subject to $g(x, y) = x^2 + y^2 - 1$. This leads to:

$$\begin{cases} y = \lambda 2x \\ x = 2\lambda y \\ x^2 + y^2 - 1 = 0. \end{cases}$$

If $y \neq 0$ then you have $\lambda = \pm \frac{1}{2}$. Then $y = x$ and, using the constrain you have $x = \pm \frac{\sqrt{2}}{2}$ and $y = \pm \frac{\sqrt{2}}{2}$. The other solution correspond to $\lambda = -\frac{1}{2}$. See Figure 1.

3 Supported vector machines: SVM

In this section we briefly introduce the theory behing SVM. This is not a deep jump into theory but just some intuitions.

3.1 Linear SVM: hard-margin

Assume you have n points $\{x_i\}_{i=1}^n$ in $\mathbb{R}^n$ and that they are labeled with labes $y_i = \pm 1$. Assume that they can be linearly separable, i.e. that there exists a hyperplane separating them. The goal is to select two parallel planes such that the distance between them is maximized. Recall that:

$$w^T x + b = 0,$$

is the margin, namely the hyperplane separating the points. All points on one side of the margin are such that:

$$w^T x + b > 0, \quad w^T x + b < 0.$$

Once that you have selected the right w and b you have a classifier that can be used on new points.

This is the mathematical setting:

$$\begin{aligned} \pi : \quad w^T x - b &= 1, \\ \pi' : \quad w^T x - b &= -1, \end{aligned}$$

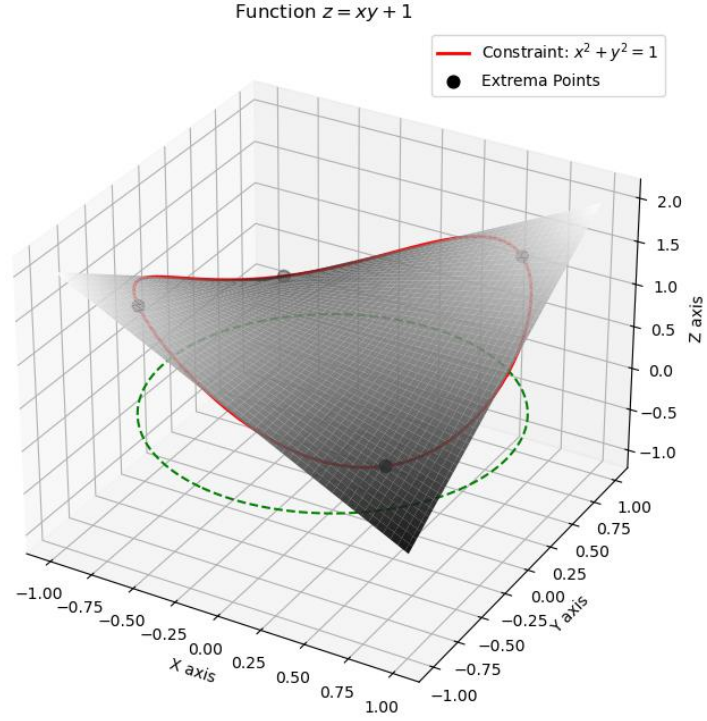


Figure 1: Example of the extrema of the function $f(x, y) = xy + 1$.

$d_{\pi, \pi'} = \frac{2}{\|w\|}$ such that $w^T x - (b + 1) \geq 0$ or $w^T x - (b - 1) \leq 0$ depending of the set of points, namely:

$$y_i = 1 \iff w^T x - (b + 1) \geq 0 \iff w^T x - b \geq 1 \iff y_i (w^T x - b) \geq 1,$$

$$y_i = -1 \iff w^T x - (b - 1) \leq 0 \iff w^T x - b \leq -1 \iff y_i (w^T x - b) \geq 1$$

and so the final problem you have to solve is:

$$\begin{aligned} & \max_{w, b} \frac{2}{\|w\|}, \\ & s.t. \\ & y_i (w^T x - b) \geq 1, \quad i = 1, \dots, n. \end{aligned}$$

The factor $b \pm 1$ is just a matter of normalization. It could be any constant instead of ± 1 .

If the points are linearly separable, by solving the maximization problem you get w and b and hence you can use the linear classifier to classify new points $\tilde{x}$.

$$w^T \tilde{x} + b > 0 \implies \tilde{y} = 1, w^T \tilde{x} + b < 0 \implies \tilde{y} = -1,$$

3.2 Linear SVM: soft-margin

Assume now that the points are not linearly separable. You can say good-bye to the hyperplane separating them. We can use the hinge function and solving the following problem:

$$\min_{w,b} \|x\|^2 + C \left[\frac{1}{n} \sum_{i=1}^n \max(0, 1 - y_i (w^T x - b)) \right],$$

where C is a penalty. Note that if the classification is “right”, then the hinge function is zero.

This problem is equivalent to:

$$\begin{aligned} \min_{w,b,\xi} \|x\|^2 + C \sum_{i=1}^n \xi_i, \\ s.t. \\ \xi_i \geq 0, \quad y_i (w^T x_i - b) \geq 1 - \xi_i, \quad i = \{1, \dots, n\}. \end{aligned}$$

4 The Dual problem

The theory behind the duality in optimization is huge and I have no time to deep dive into it. Here there are only some remarks which gives a tiny overview of the question.

Suppose you have $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}^p$, $\lambda \in \mathbb{R}^m$ and $\mu \in \mathbb{R}^p$ and that you want to solve:

$$\begin{aligned} \min f(x) \\ s.t. \quad h(x) = 0, \quad g(x) \leq 0. \end{aligned}$$

You can write the Lagrangian $L(x, \lambda, \mu) = f(x) + \lambda^T h(x) + \mu^T g(x)$. Now you can focus on the dual function $q : \mathbb{R}^{m+p} \rightarrow \mathbb{R}$ defined as:

$$q(\lambda, \mu) = \min_{x \in \mathbb{R}^n} L(x, \lambda, \mu).$$

$g(x) \leq 0$: it is violated if $g(x) > 0$ and hence $\mu^T g(x) \leq 0$: then $\mu \geq 0$ since it must be a penalty. Here there are some useful theorems.

Theorem 4.1. *Let x^* optimal solution of the primal problem. If $\lambda \in \mathbb{R}^m$, $\mu \in \mathbb{R}^p$, $\mu \geq 0$ then:*

$$q(\lambda, \mu) \leq f(x^*)$$

The dual function is a lower bound for the $f(x)$ valuated at optimal solution. So the goal is to find the greatest lower bound, namely:

$$\max_{\lambda, \mu} q(\lambda, \mu), \quad \text{s.t. } \mu \geq 0.$$

Theorem 4.2. x^* solution of the primal and (λ^*, μ^*) solution of the dual. Then

$$q(\lambda^*, \mu^*) \leq f(x^*).$$

Corollary 4.3. If $\exists x^*, \lambda^*, \mu^*$ such that $q(\lambda^*, \mu^*) = f(x^*)$ they they are optimal.

Sometimes the dual is easier to solve than the primal and hence one can solve the dual and hence getting an optimal solution for the primal. Let's see how does it works in the case of SVM problem with soft-margin.

4.1 Step by step derivation of the dual problem for SVM

The primal problem is:

$$\begin{aligned} \min_{w, b, \xi} & \|x\|^2 + C \sum_{i=1}^n \xi_i, \\ \text{s.t.} & \\ \xi_i \geq 0, & \quad y_i (w^T x_i - b) \geq 1 - \xi_i, \quad i = \{1, \dots, n\}. \end{aligned}$$

Write the Lagrangian:

$$L(w, b, \xi; \alpha, \mu) = \frac{1}{2} \|x\|^2 + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n \alpha_i [y_i (w^T x + b) - 1 + \xi_i] - \sum_{i=1}^n \mu_i \xi_i.$$

Now impose the Karush-Kuhn-Tucker conditions:

$$\begin{aligned} \frac{\partial L}{\partial w} = 0 & \implies w = \sum_{i=1}^n \alpha_i y_i x_i, \\ \frac{\partial L}{\partial b} = 0 & \implies \sum_{i=1}^n \alpha_i y_i = 0, \\ \frac{\partial L}{\partial \xi_i} = 0 & \implies 0 \leq \alpha_i \leq C. \end{aligned}$$

Plug the expression of w back into the Lagrangian and get:

$$L_D(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j.$$

and finally:

$$\begin{aligned} \max_{\alpha} & L_D(\alpha), \\ \text{s.t.} & \sum_{i=1}^n \alpha_i y_i = 0, \quad 0 \leq \alpha_i \leq C. \end{aligned}$$

Solve this problem (which is quadratic optimization), get α^* , compute $w^* = \sum_{i=1}^n \alpha_i^* y_i x_i$ and b^* from the points lie on the margin: if $0 < \alpha_j < C$ then $y_i (w^* x_i + b^*) = 1$. This last condition comes from the so called *complementary slackness*¹.

Now we show why: if $0 \leq \alpha_j \leq C$ then $y_i (w^* x_i + b^*) = 1$.

Primal feasibility $y_i (w^T x + b) \geq 1 - \xi_i$ and $\xi_i \geq 0$.

Dual feasibility $\alpha_i \geq 0$ and $\mu_i \geq 0$. The complementary slackness say:

$$\begin{aligned}\alpha_i [y_i (w^T x + b) - 1 + \xi_i] &= 0, \\ \mu_i \xi_i &= 0.\end{aligned}$$

If the constrain is active then $\alpha_i = 0$ otherwise $\alpha_i \neq 0$. Now, if

$$0 < \alpha_i < C \implies y_i (w^T x + b) - 1 + \xi_i = 0 \implies y_i (w^T x + b) = 1 - \xi_i$$

Moreover:

$$\alpha_i + \mu_i = C \implies \mu_i = C - \alpha_i > 0 \implies \mu_i > 0 \implies \mu_i \xi_i = 0 \implies \xi_i = 0.$$

Combining:

$$y_i (w^T x + b) = 1, \quad \text{for } i = 1, \dots, n.$$

Then: $b = \frac{1}{y_i} - w^T x_i = y_i - w^T x_i$ for all $0 < \alpha_i < C$. Finally, if multiple vectors x_i satisfies this you can get b^* by taking an average, increasing the stability.

Once you get w^* and b^* you have that the classifier is:

$$\begin{aligned}w^{*T} \tilde{x} + b^* > 0 &\implies \tilde{y} = 1, \\ w^{*T} \tilde{x} + b^* < 0 &\implies \tilde{y} = -1.\end{aligned}$$

5 Non-linear classifiers and the kernel-trick

We have seen that $w^T x + b = 0$ subject to $y_i (w^T x + b) \geq 1$, namely a linear classifier works only if we have linearly separable points. What if they are not? The idea is to map the original points in an higher dimensional space (by using a function) in which they can be linearly separated.

Consider $\phi : \mathbb{R}^n \rightarrow \mathcal{H}$ and write the primal problem in $\mathcal{H}$. You have to solve:

$$\min_{w, b} \frac{1}{2} \|w\|^2, \quad \text{s.t. } y_i (w^T \phi(x_i) + b) \geq 1.$$

¹I had no time to investigate this topic further

As before, you can write the Lagrangian $L(w, b, \alpha)$, imposing the KKT conditions and hence writing the dual problem:

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j \langle \phi(x_i), \phi(x_j) \rangle, \\ \alpha_i \geq 0, \quad & \sum_{i=1}^n \alpha_i y_i = 0. \end{aligned}$$

The scalar product $\langle \phi(x_i), \phi(x_j) \rangle$ might be hard to compute. Observe that, when we have to solve the optimization problem, we do not need to compute the function ϕ directly, but only the scalar product. This is the key intuition of kernel-trick.

What is the classification rule? Once you get α^* compute $w^* = \sum_{i=1}^n \alpha_i^* y_i \phi(x_i)$ and $b^* = y_i - \sum_{j=1}^n \alpha_j^* y_j \langle \phi(x_i), \phi(x_j) \rangle$ for $0 < \alpha_i < C$ (or b^* as an average of points satisfying $0 < \alpha_i < C$).

The decision function is:

$$\begin{aligned} w^{*T} \phi(\tilde{x}) + b^* &= \sum_{i=1}^n \alpha_i y_i \langle \phi(\tilde{x}), \phi(x_i) \rangle + b^* > 0 \implies \tilde{y} = 1, \\ w^{*T} \phi(\tilde{x}) + b^* &= \sum_{i=1}^n \alpha_i y_i \langle \phi(\tilde{x}), \phi(x_i) \rangle + b^* < 0 \implies \tilde{y} = -1. \end{aligned}$$

If you are able to write the scalar product $\langle \phi(x_i), \phi(x_j) \rangle$ as a function $\langle \phi(x), \phi(y) \rangle = K(x, y)$ then the problem can be highly simplified from a numerical point of view.

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j K(x_i, x_j), \\ \alpha_i \geq 0, \quad & \sum_{i=1}^n \alpha_i y_i = 0. \end{aligned}$$

Example 5.1. Assume points are as in Figure 2. Clearly they cannot be linearly separated by an hyper-plane in $\mathbb{R}^2$, but what about if we map them (in a clever way) in $\mathbb{R}^3$? Suppose to use the following function:

$$\begin{aligned} \phi : \mathbb{R}^2 &\rightarrow \mathbb{R}^3, \\ (x, y) &\mapsto (x, y, x^2 + y^2). \end{aligned}$$

Given $z_1, z_2 \in \mathbb{R}^2$ we can show that:

$$\langle \phi(z_1), \phi(z_2) \rangle = z_1^T z_2 + \|z_1\|^2 \|z_2\|^2 = K(z_1, z_2),$$

where $K(x, y)$ is the kernel we need to apply the kernel-trick. By solving the optimization problem we can get α^* and obtain the classification rule. The results is shown in Figure 3

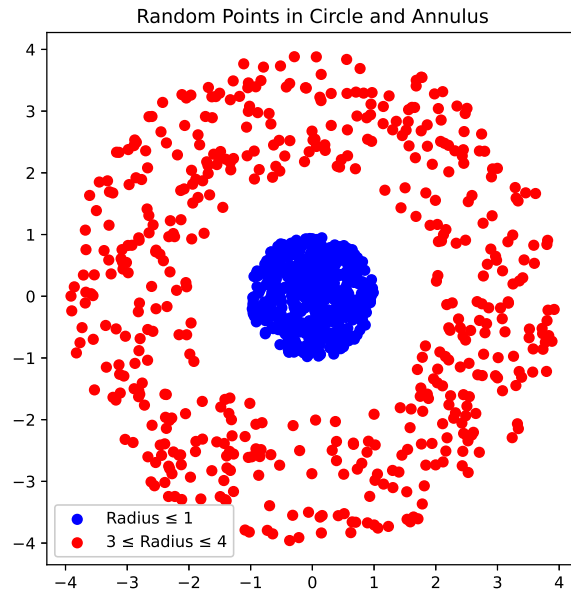


Figure 2: Points for classification.

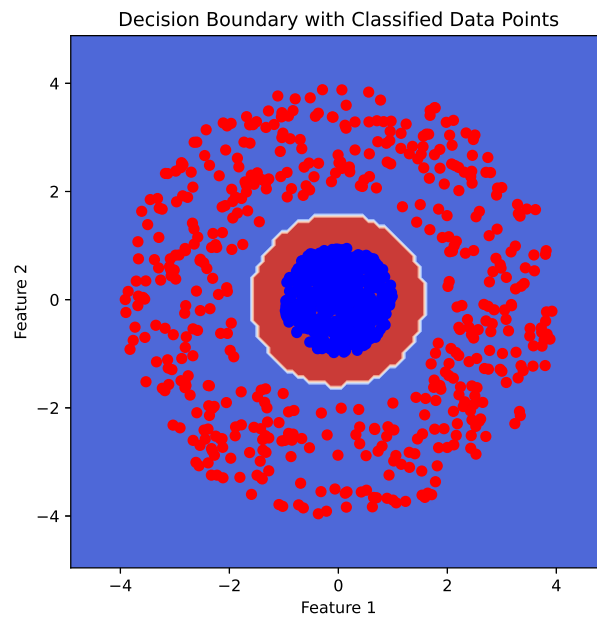


Figure 3: Decision boundary with classified data points.